

DIGITAL IMAGE PROCESSING MINI PROJECT REPORT

RESTORATION OF FINGERPRINT IMAGES FOR FORENSIC MATCHING

1. INTRODUCTION

Digital Image Processing is widely used in applications involving image enhancement, restoration, segmentation, feature extraction, and recognition. One important application is fingerprint image processing.

Fingerprints consist of unique patterns of ridges and valleys. These patterns contain important characteristics that can be used for biometric identification. Fingerprint recognition is used in security systems, authentication systems, access control, law enforcement, and forensic applications.

The quality of a fingerprint image can significantly affect the accuracy of fingerprint analysis. Fingerprint images captured under poor conditions may contain blur, noise, low contrast, broken ridges, and unwanted background regions. Such problems make it difficult to identify the ridge structures and extract reliable minutiae.

Digital Image Processing techniques can be applied to improve the quality of these images. Filtering techniques can reduce noise, contrast enhancement can improve ridge visibility, thresholding can separate the fingerprint from the background, and morphological operations can remove unwanted structures. Skeletonization can then convert fingerprint ridges into thin structures that are suitable for minutiae extraction.

In this project, a complete fingerprint processing pipeline was developed using Python. The pipeline includes image degradation, restoration, enhancement, segmentation, morphological processing, skeletonization, minutiae extraction, and experimental matching.

The project follows the application-oriented nature of the Digital Image Processing mini-project guidelines, which require a real-world imaging problem, suitable image data, restoration and/or segmentation techniques, implementation, comparison, evaluation, limitations, and future improvements.

2. PROBLEM STATEMENT

Fingerprint images may become degraded due to noise, blur, poor contrast, and other acquisition problems. Degraded images can make fingerprint ridge structures difficult to identify and can reduce the reliability of feature extraction.

The problem addressed in this project is:

To develop a Digital Image Processing pipeline that restores degraded fingerprint images, improves their ridge structures, segments the fingerprint region, extracts important minutiae, and investigates their use for fingerprint matching.

The project specifically investigates how classical image restoration and segmentation techniques can improve the quality and structural representation of fingerprint images.

3. OBJECTIVES

The main objectives of this project are:

1. To obtain and analyze a fingerprint image dataset.
2. To simulate degradation in fingerprint images.
3. To investigate noise-removal techniques.
4. To compare Median Filtering and Gaussian Filtering.
5. To investigate Gabor Filtering for fingerprint ridge enhancement.
6. To enhance local fingerprint contrast using CLAHE.
7. To quantitatively compare restoration methods using MSE, PSNR, and SSIM.
8. To segment fingerprint ridge structures using adaptive thresholding.
9. To apply morphological operations to improve the segmented image.
10. To remove small unwanted connected regions.

11. To skeletonize fingerprint ridge structures.
12. To extract ridge endings and bifurcations using the Crossing Number method.
13. To process multiple fingerprint samples using the same pipeline.
14. To perform an experimental minutiae matching procedure.
15. To analyze the failure cases and limitations of the matching method.
16. To identify possible improvements for a more reliable fingerprint-processing system.

4. DATASET / IMAGE DESCRIPTION

4.1 Dataset Source

The fingerprint images used in this project were obtained from the publicly available GitHub repository:

Minutiae-Extraction-and-Matching

Repository owner: **YogeshMoun**

The repository was cloned into the Google Colab environment and the fingerprint images were loaded using Python and OpenCV.

The project implementation uses the selected repository for the working fingerprint dataset.

4.2 Number of Images

The selected dataset contains:

170 TIFF fingerprint images

Some example image names include:

- 101_1.tif
- 101_2.tif
- 102_1.tif
- 102_2.tif
- 103_5.tif
- 103_6.tif

4.3 Image Characteristics

The dataset consists of grayscale fingerprint images containing ridge and valley patterns.

The images are suitable for:

- Fingerprint enhancement
- Noise removal
- Segmentation
- Skeletonization
- Minutiae extraction
- Experimental matching

4.4 Preprocessing

The project used several preprocessing and processing operations:

- Grayscale image processing
- Image normalization where required
- Gaussian blur
- Gaussian noise simulation
- Contrast reduction
- Median filtering
- CLAHE enhancement
- Adaptive thresholding
- Morphological processing
- Small-object removal
- Skeletonization

4.5 Dataset Limitation

The dataset consists of ordinary fingerprint images available from the selected GitHub repository. It is not specifically a dataset of real crime-scene latent fingerprints.

Therefore, the degradation used in the restoration experiment was simulated rather than obtained from actual forensic acquisition conditions.

This is an important limitation of the project.

5. METHODOLOGY

The overall methodology followed in the project is:

Input Fingerprint → Simulated Degradation → Filtering → Contrast Enhancement → Segmentation → Morphological Processing → Small Object Removal → Skeletonization → Minutiae Extraction → Experimental Matching → Evaluation

Step 1: Input Image

A grayscale fingerprint image from the dataset was selected as the input.

Step 2: Simulated Degradation

Gaussian blur, Gaussian noise, and contrast reduction were applied to create a degraded fingerprint image.

A 7×7 **Gaussian blur** was used, followed by Gaussian noise and contrast reduction.

Step 3: Noise Removal

Two spatial filtering techniques were investigated:

- Median Filtering
- Gaussian Filtering

A 5×5 **Median Filter** and 5×5 **Gaussian Filter** were compared.

Step 4: Gabor Filtering

Gabor filtering was investigated for fingerprint ridge enhancement. A fixed-orientation Gabor filter was tested.

The result produced unwanted artifacts, so it was treated as a comparison/failure case rather than the final restoration technique.

Step 5: Contrast Enhancement

Median filtering was combined with **CLAHE (Contrast Limited Adaptive Histogram Equalization)**.

The CLAHE parameters used were:

- Clip Limit = 2.0
- Tile Grid Size = 8×8

This combination was selected as the final restoration method because it produced the best quantitative results among the tested approaches.

Step 6: Segmentation

Adaptive Gaussian thresholding was applied to convert the restored fingerprint into a binary image.

Step 7: Morphological Processing

Morphological opening and closing were applied using a **3×3 kernel** to remove unwanted structures and improve ridge continuity.

Small connected regions below **50 pixels** were removed.

Step 8: Skeletonization

The cleaned binary fingerprint was skeletonized using the `skeletonize()` function from `scikit-image`.

Step 9: Minutiae Extraction

The Crossing Number method was used to detect:

- **CN = 1 $\rightarrow$ Ridge Ending**
- **CN = 3 $\rightarrow$ Bifurcation**

Additional filtering was used to reduce unwanted minutiae.

Step 10: Experimental Matching

The extracted minutiae were compared using Euclidean distance.

A distance threshold of **10 pixels** was used to identify potential matches

6. IMPLEMENTATION

The project was implemented using **Python in Google Colab**.

Libraries Used

- OpenCV
- NumPy
- SciPy
- scikit-image
- Matplotlib
- Pandas

Digital Image Processing Techniques Used

- Gaussian blur
- Gaussian noise simulation
- Median filtering
- Gaussian filtering
- Gabor filtering
- CLAHE
- Adaptive thresholding
- Morphological opening
- Morphological closing
- Small-object removal
- Skeletonization
- Crossing Number minutiae extraction
- Euclidean-distance-based experimental matching

The final restoration and segmentation implementation applies CLAHE, filtering, adaptive thresholding, morphological processing, small-object removal, and skeletonization.

The project also implemented multiple experiments to compare techniques before selecting the final approach.

7. RESULTS

7.1 Restoration Results

The restoration methods were evaluated on fingerprint **101_1** using MSE, PSNR, and SSIM.

Method	MSE	PSNR	SSIM
Degraded Image	2725.07	13.78 dB	0.5751
Gabor Filtering	13084.59	6.96 dB	0.4123
Median + CLAHE	2099.11	14.91 dB	0.7276

The values were calculated in the project implementation and comparison graphs were also generated.

Result Analysis

Median + CLAHE provided the best result because it achieved:

- Lowest MSE
- Highest PSNR
- Highest SSIM

Therefore, **Median + CLAHE** was selected as the final restoration method.

The Gabor method performed poorly in this experiment because a fixed orientation was used, while fingerprint ridges have different local orientations.

7.2 Minutiae Extraction Results

The final pipeline was applied to three fingerprint images.

Fingerprint	Ridge Endings	Bifurcations	Total Minutiae
101_1	109	14	123
101_2	72	3	75
102_1	159	18	177

The final pipeline was applied consistently to the different fingerprint samples.

7.3 Experimental Matching Results

Same Finger: 101_1 vs 101_2

Feature	Matches	Score
Ridge Endings	22	20.18%
Bifurcations	0	0.00%

Different Finger: 101_1 vs 102_1

Feature	Matches	Score
Ridge Endings	51	32.08%
Bifurcations	1	5.56%

The matching algorithm used a **10-pixel distance threshold**.

Matching Analysis

The different-finger comparison produced a higher ridge-ending score than the same-finger comparison. Therefore, the simple nearest-neighbor matching method cannot reliably distinguish matching and non-matching fingerprints.

This was treated as a limitation and failure case rather than claiming successful biometric recognition.

7.4 Visual Results

The following figures should be included in the report:

Figure 1: Original fingerprint

Figure 2: Original vs degraded fingerprint

Figure 3: Median Filtering vs Gaussian Filtering

Figure 4: Gabor ridge enhancement

Figure 5: Original → Degraded → Median + CLAHE

Figure 6: MSE, PSNR and SSIM comparison graphs

Figure 7: Binary segmentation

Figure 8: Skeletonized fingerprint

Figure 9: Extracted ridge endings and bifurcations

Figure 10: Complete pipeline:

Original → Degraded → Restored → Segmentation → Skeleton → Minutiae

The source code generates the final end-to-end visualization and minutiae visualization.

8. LIMITATIONS

The main limitations of the project are:

1. **Simulated degradation:** The blur, noise, and contrast degradation were artificially generated and may not represent all real-world forensic fingerprint conditions.
2. **Dataset limitation:** The selected dataset contains ordinary fingerprint images rather than a dedicated real latent-fingerprint forensic dataset.
3. **Limited quantitative evaluation:** Restoration metrics were evaluated on the selected sample rather than performing a comprehensive statistical evaluation across all 170 images.
4. **No segmentation ground truth:** Ground-truth segmentation masks were not available, so IoU and Dice scores were not calculated.
5. **Parameter sensitivity:** The results depend on parameters such as filter size, threshold size, morphology settings, and minutiae distance threshold.
6. **False minutiae:** Noise and imperfect segmentation may produce incorrect ridge endings or bifurcations.

7. **Simple matching algorithm:** The implemented matcher uses basic spatial distance and does not consider rotation, scale, translation, minutiae orientation, ridge frequency, or local descriptors.
8. **Matching reliability:** The experimental matching results did not reliably distinguish same-finger and different-finger samples.

Therefore, the developed system should be considered an **academic DIP prototype**, not a forensic-grade fingerprint identification system.

9. FUTURE SCOPE

The project can be improved in the following ways:

1. Use a larger and more diverse fingerprint dataset.
2. Use real latent fingerprint images for forensic applications.
3. Implement orientation-adaptive Gabor filtering.
4. Estimate local fingerprint ridge frequency.
5. Improve fingerprint region-of-interest detection.
6. Develop better minutiae validation methods.
7. Include minutiae orientation and local ridge descriptors.
8. Use robust image registration for rotation, scale, and translation.
9. Investigate RANSAC or affine transformation for alignment.
10. Develop a more advanced minutiae-based matching algorithm.
11. Evaluate the system using FAR, FRR, EER, and ROC curves.
12. Use ground-truth segmentation and minutiae data for quantitative evaluation.
13. Investigate deep-learning-based fingerprint restoration and segmentation.
14. Develop the system as a web, desktop, or mobile application.
15. Explore real-time fingerprint processing.

10. CONCLUSION

This project developed a Digital Image Processing pipeline for fingerprint image restoration, segmentation, skeletonization, minutiae extraction, and experimental matching.

A dataset of **170 TIFF fingerprint images** was used. Controlled degradation was introduced using Gaussian blur, Gaussian noise, and contrast reduction. Multiple restoration techniques were investigated, including Median Filtering, Gaussian Filtering, Gabor Filtering, and CLAHE.

Among the tested methods, **Median Filtering + CLAHE** produced the best restoration results for the evaluated sample, achieving an MSE of **2099.11**, PSNR of **14.91 dB**, and SSIM of **0.7276**.

The restored fingerprint was successfully processed using adaptive thresholding, morphological operations, small-object removal, and skeletonization. Ridge endings and bifurcations were then extracted using the Crossing Number method.

An experimental minutiae matching technique was also implemented. However, the results showed that the simple spatial matching method was not sufficiently reliable for distinguishing same-finger and different-finger images. This highlighted the importance of robust fingerprint registration and advanced feature-based matching.

The project provided practical experience in applying Digital Image Processing techniques to a real-world biometric problem and demonstrated the importance of **restoration, segmentation, feature extraction, quantitative evaluation, comparison, and failure analysis** in developing an image-processing system.